

Object Detection

A concise computer-vision reference

1. Overview

Object detection identifies instances of objects in an image and predicts a bounding box and class label for each detected instance.

2. Main Stages

A typical detector extracts visual features, predicts candidate object locations and class probabilities, and filters overlapping detections using confidence thresholds and non-maximum suppression.

3. Common Architectures

One-stage detectors such as YOLO prioritize speed, while two-stage detectors such as Faster R-CNN use a proposal stage followed by refined classification and localization.

4. Evaluation

Intersection over Union (IoU) measures overlap between predicted and ground-truth boxes. Precision, recall, Average Precision (AP), and mean Average Precision (mAP) are common evaluation metrics.

5. Applications

Applications include surveillance, autonomous systems, industrial inspection, retail analytics, medical-image analysis, and traffic monitoring.

Metric	Meaning
IoU	Overlap between predicted and ground-truth bounding boxes.
Precision	Fraction of predicted detections that are correct.
Recall	Fraction of relevant objects that are successfully detected.
mAP	Aggregated detection accuracy across classes and recall levels.